
DESI Evidence Review Packet

Manuscript, reproduction, and verification guide

Bobby Morong · Independent researcher · Prepared 2026-08-09

Independent review packet • August 2026

REVIEW STATUS

Independent error-finding and numerical verification requested. No endorsement is claimed.

Covers numerical reproduction, statistical calibration, domain framing, manuscript scope, and explicit open gates.

Prepared from the reconciled manuscript and released evidence manifest. External review remains pending.

Review purpose

This packet supports independent evaluation of “Anatomy of a $2-3\sigma$ Hint,” an implementation and stress test of published DESI DR2 compressed likelihoods. The central question is whether the reported numerical results, statistical calibrations, and interpretation are internally correct and appropriately bounded.

The work is not a raw-data reanalysis, and an independent human clean-room execution remains pending. AI-assisted software and manuscript checks are disclosed but are not treated as scientific peer review.

Requested review lanes

Lane	Best-fit reviewer	Requested output
Numerical reproduction	Scientific software developer or computational cosmologist	Fresh-environment run record, hashes, mismatches, convergence notes
Statistical review	Statistician or cosmologist familiar with likelihood-ratio calibration	Assessment of null construction, selection conditioning, intervals, and interpretation
Domain review	BAO/CMB/SN cosmologist	Scope, data provenance, comparison framing, and missing native-likelihood checks
Manuscript review	Technical scientific editor or journal-experienced researcher	Errors, ambiguous claims, missing citations, and presentation recommendations

Headline evidence ledger

Question	Released result	Interpretation boundary
Global CPL preference	$\Delta\chi^2 = -8.4575$; 70/5,000 Λ CDM mocks, $p = 0.0140$	A calibrated hint, not a discovery
Selected tracer influence	LRG2, $M = 4.1249$; 78/5,000, $p = 0.0156$	Describes localization after selection
Conditional localization	34/70, $p = 0.486$	Localization is not independent extra evidence
Direct time variation	$T = 7.8506$; 30/5,000, $p = 0.0060$	CPL-specific; not arbitrary time dependence
Direct test without LRG2	$T = 3.4263$; 349/5,000, $p = 0.0698$	Most strength is LRG2-sensitive
Held-out LRG2	joint $p = 0.0408$; D_V $p = 0.0204$; D_M/D_H $p = 0.3744$	Targeted diagnostic, not an independent discovery test
Full-CMB variant	$\Delta\chi^2 = -10.35$ with LRG2; -3.76 without	PR3-native comparison stack, not DESI's exact PR4/ACT stack

Reproduction sequence

1. Record the repository commit, platform, Python version, package versions, UTC start time, and clean Git status.
2. Follow `cleanroom/README.md` without reading expected result files.
3. Run the deterministic 20-mock smoke test twice and compare hashes.
4. Run the 5,000-mock flagship calibration and complete `cleanroom/VERIFIER_WORKSHEET.md` before opening the expected results.
5. Run the frontier package for the full/no-LRG2 direct tests and the held-out LRG2 diagnostic.

- Report every mismatch, convergence failure, platform-dependent last-bit difference, changed input, or changed command.

Files to inspect

Item	Purpose
preprint/manuscript.pdf	Journal-style reading copy
preprint/manuscript.tex	REVTeX source
paper/CLAIM_EVIDENCE_MATRIX.md	Claim-to-artifact reconciliation
RESULTS_MANIFEST.json	Canonical completion states and hashes
cleanroom/README.md	Blind execution protocol
cleanroom/VERIFIER_WORKSHEET.md	Pre-comparison result record and attestation
results/*.json	Released numerical evidence

High-value review questions

- Is the global null generated and fitted consistently with the observed statistic?
- Is conditioning on global preference strength an appropriate way to prevent double-counting localization?
- Are the direct Λ CDM-versus-CPL mocks fitted at the correct observed constant- w null in each branch?
- Does the held-out LRG2 calculation propagate parameter uncertainty correctly, and is its targeted status stated strongly enough?
- Are the DES5Y offset results framed as degeneracy/sensitivity rather than diagnosis?
- Are PR3-native and compressed-CMB results clearly distinguished from DESI's native PR4/ACT analysis?
- Are any manuscript claims stronger than the corresponding released artifact permits?

Known limitations and deferred material

- No independent human clean-room execution has yet been received.
- The official/native LRG2 likelihood reconstruction is open.
- The scientific analysis implements published compressed likelihoods rather than raw catalog and survey-systematics pipelines.
- A later DDM manuscript expansion was removed from this package because its cited gate document, runner, outputs, and CLASS patch are absent from the repository and manifest.

How to report

Please return a short signed record containing scope, environment, commit, commands, outputs/hashes, discrepancies, and an explicit statement of what was and was not independently checked. A successful rerun verifies execution of the released implementation; it does not by itself establish an independent clean-room reimplementations of the likelihood.